lec8.py

```python
#Tuple In Python
num=(10,20,30,40,50)
print(num)
tup=("Python","Cyber Security","Mysql")
print(tup)
info=("Sakshi",20,85.5,True)
print(info)
tup=()
print(tup)
print(type(tup))
# It is print a empty tuple like ()
num=(10,)
print(num)
print(type(num)) # Single element tuple me comma(,) improtant h

#Indexing In Tuple
lang=("Python","C++","Java","SQL")
print(lang[0])
print(lang[2])
print(lang[-1])

lang=("Python","C++","Java","SQL")
print(lang[-1])
print(lang[-2])
print(lang[-3])

col=("Red","Green","Blue","Yellow","Black")
```

```
print(col[0])
print(col[-1])
print(col[2])
print(col[-2])

#Slicing In Tuple
num=(10,20,30,40,50)
print(num[0:3])  #(10,20,30)
num=(10,20,30,40,50)
print(num[-3:])
#(30,40,50) last ending element
num=(10,20,30,40,50)
print(num[1:4]) #(20,30,40)
num=(10,20,30,40,50)
print(num[:4])
#first four element (10,20,30,40)
num=(10,20,30,40,50)
print(num[-2:])
num=(10,20,30,40,50)
print(num[::-1])  # for reverse

# Step slicing in tuple
num=(10,20,30,40,50,60)
print(num[0:6:2])   #(10,30,50)
num=(10,20,30,40,50,60)
print(num[1:6:2])  #(20,40,60)
#Reverse slicing
num=(10,20,30,40,50,60)
print(num[5:0:-2])
```

```
54
55
56 num=(10,20,30,40,50,60)
57 print(num[:4])
58 print(num[-3:])
59 print(num[1:4])
60 num=(10,20,30,40,50)
61 print(num[0:6:2])
62 num=(10,20,30,40,50,60)
63 print(num[5:0:-2])  #(60,40,20)
64
65 #Tuple Methods
66 num=(10,20,10,30,10,40)
67 print(num.count(10))  #3
68 colors=("Red","Blue","Red","Green","Red"
   ,"Blue")
69 print(colors.count("Red"))
70 print(colors.count("Blue"))
71 num=(10,20,30,40,50)
72 print(num.index(40))  #3
73 num=(10,20,30,20,40,20)
74 print(num.index(20))  #1
75 num=(10,20,10,30,20,10,40)
76 print(num.count(10)) #3
77 print(num.index(20)) #1
78 print(num.index(30))#3
79
80
81 num=(10,20,30,20,40,10,50)
82 print(num)
```

```
print(num.count(20))
print(num.index(40))


colors=("Red","Green","Blue","Yellow",
"Black","Whhite")
print(colors[0])
print(colors[-1])
print(colors[0:3])
print(colors[-2:])

num=(10,20,30,40,50,60)
print(num[0:6:2])
print(num[5:0:-2])

data=("Python",20,"Cyber",20,30,"Python"
)
print(data)
print(data.count("Python"))
print(data.index(20))
print(data[0:3])
print(data[-2:])

num=(5,10,15,20,25,30,10)
print(num.count(10))
print(num.index(20))
print(num[0:4])
print(num[-3:])
print(num[::-1])
```

```
110
111 #Tuple Packing
112 lang=("Python","Mysql","Cyber Security")
113 print(lang)
    #('Python','Mysql','Cyber Security')
114 #Tuple Unpacking
115 lang = "Python","Java","C++"
116 print(lang)
117 print(type(lang)) #('Python','Java','C+
    +') < class 'tuple'>
118 # Ye dono hi tuple h
119 lang=("Python","Java","C++")
120 a,b,c=lang
121 print(a) #Python
122 print(b) #Java
123 print(c) # C++
124
125 student=("Sakshi",20,85)
126 name,age,marks=student
127 print(name) #'Sakshi'
128 print(age) # 20
129 print(marks) #85
130
131 data=("Python",20,"Cyber Security",85)
132 a,b,c,d=data
133 print(a)  #'Python'
134 print(b)  #20
135 print(c)  # 'Cyber Security'
136 print(d)  # 85
137 print(type(b)) # < class 'int'>
```

```python
#SET IN PYTHON
num={10,20,30,40,50}
print(num)
num={10,20,10,30,20,40}
print(num)  # {10,20,30,40} It removes a duplicate values
info={"Python",20,85.5,True}
print(info)
emp=set()
print(emp)    #It's print a empty set
print(type(emp))
leng={"Python","Java","Python","C++","Java"}
print(leng)
num={5,10,15,10,20,5,25}
print(num)  #{5,10,15,20,25}
print(type(num)) #<class = 'set'>

#SET METHODS
num={10,20,30}
num.add(40)
print(num)
language={"Python","Java"}
language.add("C++")
print(language)

num={10,20,30,40}
num.remove(20)
print(num)
```

```
colors={"Red","Green","Blue","Yellow"}
colors.remove("Blue")
print(colors)
num={5,10,15,20,25}
num.remove(15)
print(num)
language={"Python","Java","C++","Mysql"}
language.remove("Java")
print(language)

num={10,20,30}
num.add(40)
print(num)
colors={"Red","Green","Blue","Yellow"}
colors.remove("Green")
print(colors)
num={10,20,30,40}
num.discard(50)
print(num)    # No error
num={10,20,30,40}
x=num.pop()
print(x)
print(num)
num={10,20,30,40,50}
num.clear()
print(num)  #set()
print(type(num))  #<class='set'>

num={10,20,30}
num.add(40)
```

```
num.remove(20)
num.discard(50)
x=num.pop()
print(x)
print(num)

A={10,20,30}
B={30,40,50}
result=A.union(B)
print(result)  #{10,20,30,40,50}

set1={"Python","Java"}
set2={"C++","Python","Sql"}
set=set1.union(set2)
print(set)  #{Pyhon,Java,C++,sql}

a={1,2,3}
b={3,4,5}
c={5,6,7}
result=a.union(b,c)
print(result)  #{1,2,3,4,5,6,7}

a={10,20,30,40}
b={30,40,50,60}
result=a.union(b)
print(result) #{10,20,30,40,50,60} sets
ki unique values

a={10,20,30}
```

```
b={20,30,40}
result=a.intersection(b)
print(result)
#{20,30} dono set ki commob values

a={"Python","Java","C++"}
b={"Java","Sql","Python"}
result=a.intersection(b)
print(result)

a={10,20,30,40}
b={30,40,50,60}
print(a.difference(b))  #{10,20}

a={10,20,30}
b={20,30,40}
print(b.difference(a))  #{40}

a={"Pyhthon","Java","C++"}
b={"Java","Sql"}
print(a.difference(b))  #{"Python",C++}

a={10,20,30,40,50}
b={20,40,60}
print(b.difference(a))
```

```
251
252
253 text=input("Enter the string: ")
254 count=0
255 for ch in text:
256     if ch in "aeiou":
257         count=count+1
258 print("Total voweles",count)
259
260
261 text=input("Enter the string: ")
262 count=0
263 for ch in text:
264     if ch in " ":
265         count=count+1
266 print("Total space",count)
267
268
269 text="Hello Python"
270 count=0
271 for ch in text:
272     if ch.isdigit():
273         count=count+1
274 print("Total digit",count)
275
276 text="Python123"
277 count=0
278 for ch in text:
279     if ch.isdigit():
```

```
            count=count+1
print("Total digit",count)

text="Hello PYTHON"
count=0
for ch in text:
    if ch.isupper():
        count=count+1
print("Total upper letters",count)


text="Python is easy"
count=0
for ch in text:
    if ch.split():
        count=count+1
print("Total words",count)
```

```
308
309
310 num=[15,45,22,60,35,10,80]
311 new_list=[]
312 for i in num:
313     if i > 30:
314         new_list.append(i)
315 print(new_list)
316
317
318 num=[8,12,15,20,25,32,40,45]
319 new_list=[]
320 for i in num:
321     if i % 4 == 0:
322         new_list.append(i)
323 print(new_list)
324
325
326 names=["Amit","Rahul","Neha","Priyanka",
    "Rohit","Anjali"]
327 new_list = []
328 for i in names:
329     if len(i) > 5:
330         new_list.append(i)
331 print(new_list)
332
333
334
335
```

```
335
336 num=[10,15,20,25,30,45,50,60,75]
337 new_list=[]
338 for i in num:
339     if i%3==0 and i%5==0:
340         new_list.append(i)
341 print(new_list)
342
343
344
345 num=[10,-5,20,-8,30,-2,5]
346 new_list=[]
347 for i in num:
348     if i < 0:
349         new_list.append(0)
350     else:
351         new_list.append(i)
352 print(new_list)
353
354 num=[12,17,20,25,30,41]
355 new_list=[]
356 for i in num:
357     if i %2==0:
358         new_list.append("Even")
359     else:
360         new_list.append("Odd")
361 print(new_list)
362
363
```

```
362
363
364
365 employees = [
366     {"name": "Amit", "salary": 55000},
367     {"name": "Rahul", "salary": 70000},
368     {"name": "Neha", "salary": 80000},
369     {"name": "Rohit", "salary": 45000}
370 ]
371 new_list=[]
372 for employee in employees:
373     if employee ["salary"] > 60000:
374         new_list.append(employee["name"
    ])
375 print(new_list)
376
377
378 employees = [
379     {"name": "Amit", "department": "IT"
    },
380     {"name": "Rahul", "department": "HR"
    },
381     {"name": "Neha", "department": "IT"
    },
382     {"name": "Rohit", "department":
    "Sales"}
383 ]
384 new_list=[]
385 for employee in employees:
```

```
    if employee["department"]=="IT":
        new_list.append(employee["name"
])
print(new_list)



students = [
    {"name": "Aman", "marks": 65},
    {"name": "Priya", "marks": 80},
    {"name": "Riya", "marks": 72},
    {"name": "Karan", "marks": 55}
]
new_list=[]
for student in students:
    if student ["marks"] >= 70:
        new_list.append(student["name"])
print(new_list)



employees = [
    {"name": "Amit", "department": "IT"
, "salary": 55000},
    {"name": "Rahul", "department": "HR"
```

```
409     {"name": "Neha", "department": "IT"
    , "salary": 80000},
410     {"name": "Rohit", "department": "IT"
    , "salary": 50000}
411 ]
412 new_list=[]
413 for employee in employees:
414     if employee["salary"] >=60000 and
    employee ["department"] == "IT":
415         new_list.append(employee["name"
    ])
416 print(new_list)
417
418
419 students = [
420     {"name": "Aman", "marks": 75,
    "stream": "Science"},
421     {"name": "Priya", "marks": 85,
    "stream": "Arts"},
422     {"name": "Riya", "marks": 70,
    "stream": "Science"},
423     {"name": "Karan", "marks": 90,
    "stream": "Commerce"}
```

```
]
new_list=[]
for student in students:
    if student["marks"] >= 70 and student["stream"] == "Science":
        new_list.append(student["name"])
print(new_list)




employees = [
    {"name": "Amit", "salary": 45000, "department": "IT"},
    {"name": "Rahul", "salary": 60000, "department": "HR"},
    {"name": "Neha", "salary": 48000, "department": "IT"},
    {"name": "Rohit", "salary": 55000, "department": "Sales"}
]
new_list=[]
for employee in employees:
    if employee["salary"] > 50000 or employee["department"] == "IT":
        new_list.append(employee["name"])
print(new_list)
```

```
446
447 students = [
448     {"name": "Aman", "marks": 75,
"stream": "Science"},
449     {"name": "Priya", "marks": 80,
"stream": "Science"},
450     {"name": "Riya", "marks": 55,
"stream": "Science"},
451     {"name": "Karan", "marks": 70,
"stream": "Commerce"},
452     {"name": "Neha", "marks": 60,
"stream": "Science"}
453 ]
454 new_list=[]
455 for student in students:
456     if student ["marks"] > 60 and
```

```
student["marks"] <= 80 and student [
"stream"] == "Science":
457             new_list.append(student["name"])
458 print(new_list)
459
460
461 employees = [
462     {"name": "Amit", "salary": 45000,
"department": "IT"},
463     {"name": "Rahul", "salary": 60000,
"department": "HR"},
464     {"name": "Neha", "salary": 48000,
"department": "IT"},
465     {"name": "Rohit", "salary": 55000,
"department": "Sales"},
466     {"name": "Priya", "salary": 40000,
"department": "HR"}
467 ]
468 new_list=[]
469 for employee in employees:
470     if employee ["salary"] < 50000 or
employee ["department"] =="HR":
471             new_list.append(employee["name"
])
472 print(new_list)
```